

Final evidence scoreboard — 8-gene RAI differentiation panel

All headline statistics across discovery + validation + mechanistic + clinical anchors

	Cross-cohort meta forest (n = 6 cohorts) <i>pooled Cohen d</i>	-2.48 [-2.64, -2.33]
	Continuous panel z PFI Cox (TCGA, n = 477) <i>HR per unit</i>	0.60 [0.42, 0.87], p = 0.007
	Mun 2025 proteome ATC vs PTC dark-matter <i>Fisher OR</i>	8.54, p = 2.7 × 10 ⁻¹⁵
	Lee 2024 Korean histology dose-response <i>KW p-value</i>	p = 8.7 × 10 ⁻⁵⁵ (n = 632)
	Mutation group × panel z (TCGA) <i>KW p-value</i>	p = 9.5 × 10 ⁻³⁷ (n = 500)
	AJCC stage × panel z (TCGA, I → IV) <i>KW p-value</i>	p = 2.4 × 10 ⁻⁴
	dark-matter · TERT+ PFI rate <i>rate vs ref</i>	27.8% vs 7.0% (4×, log-rank p = 0.037)
	Lu 2023 sc-RNA per-cell distribution <i>KW H-stat</i>	H = 10,471.5, p < 10 ⁻³⁰⁰ (cells = 14,624)
	Top-quartile purity zone separation <i>KW p-value</i>	p = 1.0 × 10 ⁻²¹ (n = 131)
	8-gene Spearman ρ (TCGA cross-correlations) <i>mean ρ</i>	all pairs positive, max 0.86 (TG ↔ TPO)
	Cherry-pick null permutation (n = 1,000) <i>empirical p</i>	p = 0.014 (matched-variance null)
	Methylation × RNA per-gene correlation <i>Δβ vs Δrna</i>	8/8 panel genes show β↑ + RNA↓